COUNTER-DRONE TECHNIQUES
To Detect And Neutralise Hostile Drones

Drone and anti-drone technologies will continue to evolve and co-exist. The next generation of unmanned aerial vehicles (UAVs) will be lighter, smaller, more complex, and be able to multi-task. Hence the necessity to figure out new, more effective ways of shooting these platforms down

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An insect-like microdrone (Source: https://dronedj.com)

The use of drones or unmanned aerial vehicles (UAVs), both for military and civilian purposes, has been increasing. At the same time, counter-drone systems have also been developing to address their rising threat. The drones are now available from nano insect-size at one end to a large size comparable to a medium-sized passenger jet aircraft on the other end.

The availability of advanced navigation and satellite communication technologies has revolutionised the use of drones, making them more versatile. They can now endure missions for 30 to 40 hours, far beyond the capabilities of human crew. With in-flight refuelling and ultra-efficient solar power, the drones can provide an even greater range of operation.

Hostile drones pose threat to military and strategic installations and public security. One way to engage an enemy with minimum casualties is using drones that do not carry human operators, use aerodynamic forces for lift, fly autonomously or are piloted remotely, are either expendable or recoverable, and can carry both lethal and non-lethal payloads.

Drones are becoming the preferred means for intelligence gathering, surveillance, reconnaissance, electronic warfare, strike missions, air combat, and search and rescue missions due to their capability to loiter, search, identify, and strike targets while minimising the